B.Tech Degree VIII Semester Examination April 2012

ME 802 MACHINE DESIGN II
(2002 Scheme)

Time: 3 Hours Maximum Marks: 100

(Design data book is permitted. Assume reasonable data wherever necessary)

I. (a) Find a relation for the frictional torque acting on a centrifugal clutch. (5)

(b) Define (i) self energizing brake (ii) self locking brake. (5)

(c) A single plate clutch with both sides effective, is required to transmit 25KW at 900 rpm. The outer diameter of the plate is 350mm. The maximum intensity of pressure over the friction surface is not to exceed 0.1N/mm^2 (1 bar). Considering uniform wear criterion and assuming coefficient of friction as 0.25, determine (i) the inner diameter of the plate, and (ii) axial force required to engage the clutch. (15)

OR

II. (a) Derive the relation for the ratio between belt tensions of flat belt drive. (5)

(b) Explain the chordal action chains. (5)

(c) A V-belt drive is to transmit 15KW to a compressor. The motor runs at 1150 rpm and the compressor is to run at 400 r.p.m. Determine:
(i) belt specifications
(ii) number of belts
(iii) correct center distance
(iv) drive pulley diameters (15)

III. (a) State and explain law of gearing. (5)

(b) Define the terms: (i) pitch circle (ii) pitch point (iii) circular pitch (iv) backlash (v) module. (5)

(c) Design a pair of helical gears to transmit 30KW at a speed reduction 4 : 1. The input shaft rotates at 2000 rpm. The helix and pressure angles equal to 25° and 20° respectively. The number of teeth on the pinion may be taken as 30. (15)

OR

IV. (a) Explain the term interference in involute profiles. (5)

(b) Make a comparison of cycloidal and involute tooth forms. (5)

(c) Design the teeth of a pair of bevel gears to transmit 18.75KW at 600rpm of the pinion. The velocity ratio should be about 3 and the pinion should have about 20 teeth which are full depth, 20° involute. Find the module, face width, diameter of the gears, and pitch cone angle for both gears. (15)

V. (a) What are the characteristics of a good bearing material? (5)

(b) Explain the principle of hydrodynamic lubrication. (5)

(c) The load on a journal bearing is 150kN due to turbine shaft of 300mm diameter running at 1800 rpm. Determine the following:
(i) Length of the bearing if the allowable bearing pressure 1.6 N/mm^2.
(ii) Amount of heat to be removed by the lubricant per min, if the bearing temperature is 60°C, and the viscosity of oil at 60°C is 20 centipose (0.02 Kg/m-s) and the bearing clearance is 0.25 mm. (15)

OR

VI. (a) Classify the rolling contact bearings based on (i) shape of the rolling elements (ii) nature of the loads resisted. (5)

(b) Differentiate between static and dynamic load carrying capacities of rolling contact bearings. (5)

(c) A ball bearing is required to resist a radial load of 10KN and a thrust load of 5kN. The average life of the bearing is to be 5000 hours with inner race rotation at 980 rpm. What basic dynamic load rating must be used in selecting the bearing? If the bearing is to have a life of 5000 hours at a reliability of 97%, what is the required basic dynamic load rating? (15)

VII. (a) What are the main design considerations for rolled sections? (8)

(b) What are the design considerations for castings? (7)

(c) Explain various methods of representations of surface finish in working drawing. (5)

(d) Explain the steps involved in the preparation of working drawings. (5)

OR

VIII. (a) What are the general design considerations for parts produced on milling machines? (10)

(b) Explain the design considerations for welded joints. (10)

(c) How surface roughness is measured? (5)

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